



Fig. 5: Delta VFD-M

micro AC drive. Its compact design is ideal for small- and medium-horsepower applications. M drive is designed to provide an ultra-low-noise operation and includes several

innovative technologies that reduce interference.

This drive can have numerous applications like a packing machine, dumpling machine, treadmill, temperature/humidity-control fan for agriculture and aquaculture, mixer for food processing, grinding machine, drilling machine, small-size hydraulic lathe, elevator, coating equipment, small-size milling machine, robot arm of an injection machine (clamp), wood machine (two-side woodworking planer), edge-bending machine, elasticiser and so on.

Steps for complete motor control

1. Check connections of L1, L2, L3; T1, T2, T3 (used for giving 3-phase input to VFD and connecting motor to it) and wires coming out of M0, M1 and GND.

2. Switch on the 3-phase power supply.

3. For programming the VFD-M:

- (i) Press Mode
- (ii) On F60.0, press Enter
- (iii) Click Mode
- (iv) Select the appropriate parameter using Up/Down keys available on the keypad.
- (v) For example, for Pr0, select P 00.
- (vi) According to the manual, set the parameters for required mode of operation
- (vii) Press Enter→End

Note: At any point, press Mode to go to the previous step.

For motor running from external controls, we have three

modes of operation; two are 2-wire method and one is 3-wire method. Other than these, there is the default method operable from the digital keypad.

First, carry out a trial run to check all connections.

Test run for VFD

The factory setting of the operation source is from the digital keypad (Pr.01 = 00). You can perform a trial run by using the digital keypad with the following steps:

1. After applying power, verify that the display shows F60.0Hz. When AC motor drive is in standby, STOP LED and FWD LED will light up.
2. Press key to set frequency to 5Hz.
3. Press key. RUN LED and FWD LED will light up, which indicates operation command is forward running. And if you want to change to reverse running, you should press . And if you want to decelerate to stop, please press .

VFD-M programming

There are two contacts, M0 and M1. Whenever M0 is closed, VFD goes into run mode. If it is open, there is no rotation of the motor. M1 decides the direction of rotation. If M1 is open, it rotates in forward direction; if closed, in reverse direction.

Parameters for the above mode are set as:

Pr.00 set to 01 (for controlling master frequency via potentiometer)

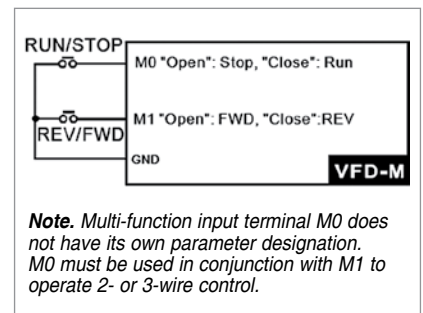


Fig. 8: Two-wire operation: Only Pr.38 can be set to "1"

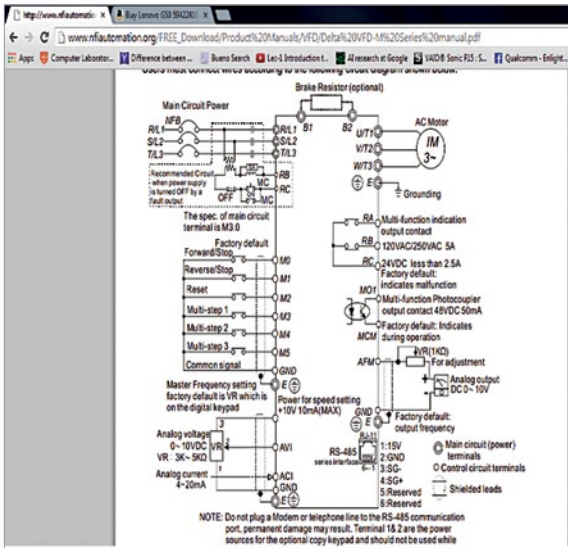


Fig. 6: Pin-out connection diagram for VFD-M

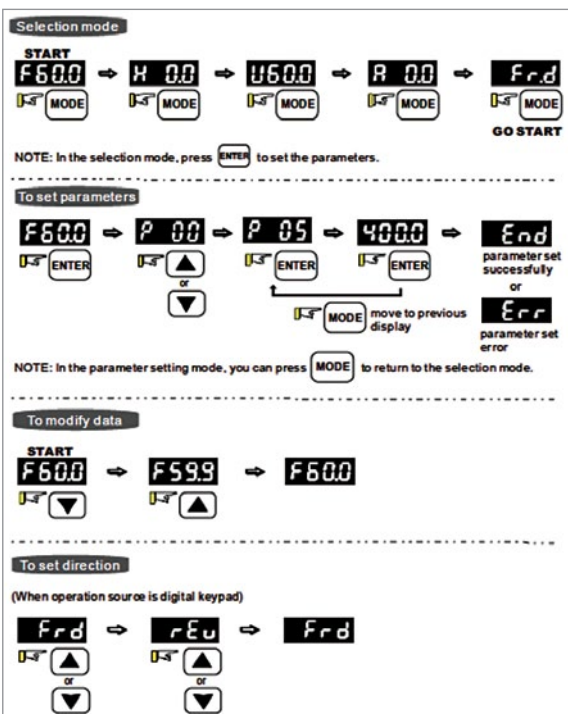


Fig. 7: Controlling a digital keypad on Delta VFD-M